

In vivo monosomy 3 detection of posterior uveal melanoma: 3-year follow-up

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Abstract

Background Monosomy 3 is a highly specific marker for poor prognosis in posterior uveal melanoma. Unfortunately, cytogenetic prognostication is limited to enucleated eyes or resected tumors. The aim of this study was to evaluate mid-term natural history and safety of in vivo detection of chromosome 3 status in posterior uveal melanomas undergoing plaque brachytherapy.

Methods A 25-gauge transscleral fine needle aspiration biopsy (FNAB) was performed in 32 eyes affected by posterior uveal melanoma undergoing plaque brachytherapy, just before applying the radioactive plaque. Sampled material underwent fluorescence in situ hybridization (FISH) with centromeric probes for chromosome 3. All patients had a follow-up of at least 36 months.

Results Mean follow-up was 47.1 ± 8.5 months. Mean largest basal diameter and mean thickness of the tumors were 12.5 ± 2.7 mm and 8 ± 2.3 mm respectively. FNAB yielded sufficient material in 26 of 32 cases (81.2%). Adequacy of the sample ranged from 91.1% (ciliary body tumors) to 76.8% (choroidal tumors). Seventeen cases had monosomy 3 (65.3%). No correlation was found between monosomy 3 and tumor dimensions or location (ciliary body vs choroidal tumors). No early and mid-term local complications were documented. Seven patients (21.8%) died during follow-up: five (15.6%) of them died due to metastatic disease (all had monosomy 3 tumors).

Conclusions Posterior uveal melanomas may be adequately and safely sampled, by intra-operative transscleral FNAB, to detect in vivo monosomy 3.

Keywords Uveal melanoma · Fine needle aspiration biopsy · FISH · Monosomy 3 · Prognosis

The authors have no financial interest in the subject of this paper.

The authors have full control of all primary data and they agree to allow Graefe's Archive for Clinical and Experimental Ophthalmology to review their data if requested.

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Introduction

Posterior uveal melanoma is the most common primary intraocular tumor in adults with an annual incidence of five to seven cases per million population [1]. Overall mortality remains high (about 50%), although less than 2% of affected patients have clinically detectable metastasis at presentation [2]. After clinical diagnosis of hepatic metastases, life expectancy is extremely poor, and mean survival time is about 7 months [3]. Since most small and medium posterior (choroid and ciliary body) uveal melanomas are currently conservatively treated with radiation, tissue is usually not available for cytologic or histologic evaluation [4, 5]. After any conservative radiation treatment, prognosis for survival is currently based only on clinical information,

but the real predictive value of these parameters has recently been questioned [6–8].

Some retrospective analyses of cytogenetic data have hypothesized that two cytogenetic pathways of clonal evolution exist in uveal melanoma: one starts with the loss of an entire chromosome 3 and continues with gains of 8q; the second pathway starts with gain of 6p [9–11]. Long-term studies have shown that about 70% of patients with monosomy 3 in the primary tumor died from metastases within 4 years after the initial diagnosis, whereas tumors with normal chromosome 3 status (disomy 3) rarely gave rise to metastatic disease [12, 13]. Therefore, monosomy 3 is considered a highly specific marker for poor prognosis in posterior uveal melanoma [6, 8, 14–16]. Currently, cytogenetic studies on posterior uveal melanomas are performed only on biologic material taken from enucleated eyes or histologic specimens from resected tumors [16, 17]. Moreover, accurate identification of patients at high risk of developing metastatic disease may be relevant for future clinical management, but patients treated with radiation seem to be excluded from any prognostic information, except clinical data [6–8].

The aim of this study was to evaluate, in a consecutive series of patients, accuracy and medium-term local safety of the in vivo detection of monosomy 3 in posterior medium-large uveal melanomas undergoing plaque brachytherapy. The cytologic sampling was performed during the brachytherapy surgical procedure and FISH (fluorescence in situ hybridization) analysis was applied to FNAB (fine needle aspiration biopsy) specimens. Patients 1 to 7 had previously been included in a short report discussing the feasibility of this technique [18].

Material and methods

This study complied with the tenets of the Declaration of Helsinki, and was approved by the IRB of our institution. Patients were recruited from those referred to the Ophthalmic Oncology Unit of the Department of Ophthalmology, University of Padova, Padova, Italy, between September 2001 and November 2003. Each patient underwent full preoperative ophthalmic and systemic examination. The ophthalmoscopic aspects of the tumors were documented by fundus photography. Melanoma dimensions were also documented using A- and B-scan ultrasonography. Liver enzymes and liver ultrasonography were used to evaluate the presence of metastatic disease at baseline. Eligible patients were affected by medium-large uveal melanoma, and were free of metastasis and other cancers at enrollment, free of melanoma extrascleral extension, 21 years or older, and free from other life-threatening coexisting systemic diseases. Informed consent was obtained after explanation

of the nature of the disease and the possible implications of this study. All patients were scheduled for standard plaque brachytherapy with I125, under general anaesthesia. Thirty-two consecutive patients undergoing plaque radiotherapy, with follow-up longer than 36 months, were included in this report.

Patients underwent an in-vivo intraocular FNAB during the procedure for radioactive plaque application, as previously described [18]. Briefly, the cytologic sampling was made through the sclera, just before suturing the active I125 plaque over the tumor base. A standard FNAB procedure was performed using a 25-gauge blunt needle, 27 mm in length (Becton Dickinson, New York, NY, USA). A 200–300 μ pre-planned scleral incision was made, and the needle inserted into the tumor through the remaining sclera. A double-pass sampling was performed. The scleral incision was sutured with 7.0 Polyglactin suture, and the radioactive plaque immediately placed over the tumor base. Tumor specimens obtained by FNAB were collected in culture medium RPMI 1640 (Euroclone Life Science, Milano, Italy).

A standard dose of 100 Gy was delivered to the tumor apex. All patients underwent a 1-, 3- and 6-month follow-up examination, and every 6 months thereafter. Follow-up examination included full ophthalmologic examination, fundus photography and A- and B-scan ultrasonography. Orbit nuclear magnetic resonance, liver enzymes analysis and liver ultrasonography were also performed every 6 months. When the site of needle aspiration was detectable through slit lamp examination, this area underwent regular photographic follow-up.

The sampled material underwent FISH for chromosome 3 detection following manufacturer's procedures as previously described [18]. After sedimentation, the material was enzymatically digested with collagenase II (Worthington Biochemical Corporation, Lakewood, NJ, USA) 1400 U/ml at 37°C for 2 hours. The suspension was then washed in RPMI 1640 and used to prepared cytopins. Slides were fixed with a cytological fixative (Bio-Fix; Bio-Optica, Milan, Italy), and stored at –20°C. FISH analysis was performed with a centromeric probe for chromosome 3 labeled with SpectrumOrange and centromeric probe for chromosome 10 labeled with SpectrumGreen (Abbott-Vysis, Downers Grove, IL, USA) following manufacturer's procedure. Slide and probe were codenatured in Hybrite (Vysis) at 75°C for 5 min and hybridized in a humid chamber overnight at 42°C. Post-hybridization washes were made at 73°C in 0.4xSSC/0.3% NP-40 for 2 min and at room temperature in 2xSSC/0.1% NP-40 for 1 min. Slides were air dried and mounted with a Vectashield mounting medium with DAPI (Vector Laboratories, Burlingame, CA, USA). Microscope analysis was carried out with a fluorescent microscope (Zeiss Axioplan fluorescent micro-

scope) equipped with a cooled charge-coupled-device (CCD) camera (Hamamatsu, Hamamatsu-city, Japan) and appropriate single band and triple band filters. Images were analyzed using CROMOFISH software (Amplimedical, Assago-MI, Italy). At least 100 cells were evaluated for each sample; loss of chromosome 3 was reported when more than 15% of cells showed a single signal for chromosome 3 (cut-off 15%).

Results

Thirty-two patients affected by posterior uveal melanoma (mean age, 63.3 ± 10.4 years; range, 36 to 82 years) are included in this report. Eighteen patients were male (56.2%) and 14 female (43.8%). The right eye was affected in 19 patients (59.4%) and the left eye in 13 cases (40.6%).

Twenty-one melanomas were purely choroidal in location (65.6%). Eleven eyes were affected by ciliary body tumors (34.4%). Sixteen tumors were located in the nasal quadrants (50%), 13 in the temporal (40.6%) and three in the posterior pole (9.4%). Mean tumor largest basal diameter was 12.5 ± 2.7 mm (range, 7–15 mm) and mean tumor thickness was 8 ± 2.3 mm (range, 4–11 mm). Mean follow-up was 47.1 ± 8.5 months (range, 37–62 months). No local complications were observed during the surgical procedure or follow-up, and no extrascleral or orbital recurrences were detected. Demographic and tumor characteristics and sampling results are summarized in Table 1.

Transscleral FNAB yielded enough material for FISH analysis in 26 of 32 cases (81.2%). In the remaining six cases the aspirate was graded as insufficient (18.8%). FNAB yielded sufficient specimens for FISH analysis in ten of 11 ciliary body tumors (91.1%) and in 16 of 21 choroidal neoplasms (76.8%). No correlation was found

Table 1 FNAB results and tumor characteristics

	Age	Eye	ch/cb	Location	LBD (mm)	Thickness (mm)	FNAB sample	Monosomy 3	% of positive cells	MTD
1	82	OD	cb	N	11	9	Sufficient	No	0%	No
2	62	OD	ch	PP	12	8	Insufficient	—		No
3	75	OD	cb	TI	15	10	Sufficient	Yes	84%	No
4	65	OD	ch	PP	8	7.5	Insufficient	—		No
5	71	OS	ch	TS	13	9	Sufficient	No	0%	No
6	82	OD	cb	TS	11	6.7	Sufficient	Yes	96%	No
7	63	OS	ch	TS	13	5.8	Sufficient	Yes	98%	No
8	68	OS	ch	TI	15	6	Insufficient	—		No
9	73	OD	cb	NS	12	7.5	Sufficient	Yes	73%	No
10	82	OD	cb	TS	12	5.5	Sufficient	Yes	94%	No
11	67	OD	ch	T	12	7.2	Sufficient	Yes	94%	Liver
12	60	OD	ch	NS	15	12.2	Sufficient	No	0%	No
13	63	OD	ch	NI	12	9	Insufficient	—		No
14	57	OS	ch	N	14	10	Sufficient	Yes	97%	Liver
15	60	OS	ch	T	12	6	Insufficient	—		No
16	60	OS	cb	NI	10	8.5	Sufficient	Yes	92%	No
17	57	OD	ch	N	13	9	Sufficient	No	0%	No
18	65	OD	cb	N	7	7.5	Insufficient	—		No
19	58	OD	ch	N	13	10	Sufficient	Yes	73%	No
20	55	OD	ch	N	13	8.2	Sufficient	Yes	83%	No
21	70	OS	ch	N	10	8	Sufficient	Yes	90%	No
22	63	OS	cb	TS	15	11	Sufficient	No	0%	No
23	75	OS	ch	N	15	11	Sufficient	No	0%	No
24	44	OD	cb	T	11	11	Sufficient	No	0%	No
25	36	OS	ch	N	9	11	Sufficient	No	0%	No
26	68	OD	ch	PP	13	4	Sufficient	No	0%	No
27	65	OS	ch	N	8	10	Sufficient	Yes	89%	Liver
28	82	OD	ch	TS	11	10	Sufficient	Yes	100%	No
29	75	OS	cb	N	15	11	Sufficient	Yes	73%	No
30	75	OD	cb	TI	15	10	Sufficient	Yes	0%	No
31	74	OS	ch	NI	15	5	Sufficient	Yes	96%	Liver
32	70	OD	ch	TI	12	5.5	Sufficient	Yes	92%	Liver

*ch=choroidal; cb=ciliary body; LBD=largest basal diameter; MTD=metastatic disease; N=nasal; T=temporal; I=inferior; S=superior; PP=posterior pole.

between tumor dimension (thickness and largest basal diameter) or tumor location (ciliary body vs choroidal location) and adequacy of FNAB ($p>0.05$). Monosomy 3 was detected in 17 cases (65.3%) and disomy 3 in the remaining nine cases (34.7%). No correlation was found between chromosome 3 status and tumor dimensions ($p>0.05$), and between monosomy 3 and tumor location (ciliary body vs choroidal location) ($p>0.05$). Among the sixteen monosomy 3 tumors, the mean percentage of monosomic cells in each sample was $89\%\pm 9.4\%$ (range, 73–100%). Seven patients (21.8%) died during follow up. Five of them, all affected by monosomy 3 melanoma, died due to liver metastatic disease, which developed 16 ± 2 months (range, 12–20 months) after treatment, without any evidence of extrascleral recurrence in the sampled site. Two patients (6.2%) died of unrelated causes.

Discussion

Cytogenetic analysis of posterior uveal melanoma has contributed to the delineation of the pattern of genetic alterations in this tumor [7, 9]. Recurrent chromosomal abnormalities affecting chromosome 1, 3, 6 and 8 have been described and correlated with prognosis [7, 19]. Loss of chromosome 3, loss of short arm of chromosome 1 and gain of 8q have been associated with decreased survival [10, 11, 14, 15]. Because histologic material is usually not available when conservative treatment is used in the management of posterior uveal melanoma, conservatively treated patients have no complete information about life prognosis [6, 9, 11]. Moreover, clinicians are unable to accurately select patients at high risk for metastasis to be included in ongoing studies of adjuvant chemotherapy [6]. These considerations led us to search for a simple and reliable technique to detect monosomy 3 in vivo.

FNAB is currently used to sample indeterminate intraocular lesions to allow for a preoperative cytologic diagnosis [20, 21]. Early and long-term complications following intraocular FNAB are rare and time-limited [20, 21]. Concerns about tumor seeding due to this diagnostic procedure have been largely dispelled by literature data [20, 21]. Only one suspected (not adequately proven) clinical case of extrascleral recurrence has been recently reported after diagnostic intraocular transscleral FNAB [22]. No treatment was applied over the sampled area and the tumor base, compared to our approach which includes full radiation treatment of the entry site.

FNAB has already been used to sample enucleated eyes to investigate cytogenetic abnormalities with promising results [2, 17]. Therefore, the application of FISH to FNAB is considered a reliable method for detecting genetic prognostic parameters, and cytologic material obtained

using FNAB seems highly representative of the tumor's genetic characteristics [2, 17]. We considered that moving detection of monosomy 3 from the ex vivo to the in vivo arena may represent a relevant improvement in the clinical identification of this risk factor in patients affected by posterior uveal melanoma.

We have recently reported the feasibility of detecting monosomy 3 in FNAB samples using FISH in a small cases series. FNAB yielded enough material in 7 of 8 eyes (87.5%) [18]. The present larger prospective series confirms that FNAB yields enough material to detect monosomy 3 in over 80% of sampled cases. The proportion of adequate biopsies for analysis was higher for ciliary body melanomas than for choroidal melanomas, without reaching statistical significance. This fact may be explained by the higher technical difficulties of sampling a more posteriorly located tumor. A transvitreal approach has recently been investigated for posteriorly located tumors, in a case series including 140 eyes with limited follow-up [23]. According to this study, the tumors posterior to the equator (67 of 140 eyes: 48%) were sampled via pars plana, yielding sufficient material in 65 of 67 cases (97%). Unfortunately, this study reports a high number of local complications (vitreal haemorrhages in 64 eyes), probably related to the direct transretinal approach to the tumor apex [23]. In the same series, a lower rate of sufficient material (75% vs 97%) was obtained when the tumors were sampled with 30- vs 27-gauge needle. Balancing needle diameter vs tumor approach needs further investigation [23]. A 25-gauge vitrector has recently been claimed to give more material than transvitreal or transscleral FNAB [24]. Unfortunately, this technique is more aggressive and surgically demanding, and its safety is unknown. Moreover, using the FNAB technique, a double-pass is always recommended to reduce the incidence of insufficient sampling. Sampled tumors were not cytologically classified after FNAB, because FNAB material was fully used for cytogenetic analysis.

FNAB adequacy, as reported by Sisley et al., seems independent of melanoma dimensions, as confirmed by our data [17]. FISH testing showed monosomy 3 in 65.3% of our cases. Previous ex vivo studies reported monosomy 3 in 55–65% of sampled cases [7, 12]. The high percentage of monosomy 3 cells in each positive sample (mean, $89\%\pm 9.4\%$) shows that, when present, monosomy 3 is fully exposed and easily detected by FNAB sampling. The presence of heterogeneous distribution of cytogenetic abnormalities, namely monosomy 3, has been recently evaluated in some case series, with conflicting results [25–27]. Maat et al. used, on enucleated eyes, some laboratory techniques which may not represent the characteristics of in vivo tumors [26]. Meir et al., who used a larger pattern of laboratory investigation, were not able to detect tumor heterogeneity [27–29].

No correlation was found between monosomy 3 and tumor dimension. Considering literature data, the existence of a correlation between these two parameters still remains unclear [8, 23, 30].

Our sampling technique for monosomy 3 detection differs from standard FNAB [21]. We use partial scleral incision, followed by suture after sampling, to avoid excessive pressure when penetrating the sclera, particularly relevant in small tumors. This maneuver, followed by direct application of radioactive plaque over the tumor base, improves the safety of tumor sampling. Mid-term safety of this sampling procedure is proven by the absence of any local extrascleral recurrence (regularly and adequately checked) with a mean follow-up of about 4 years in our population. Moreover, patients' mortality rate from liver metastasis 36 months after treatment agrees with previously published data [31].

Other laboratory techniques have been used to investigate monosomy 3 in uveal melanoma: single nucleotide polymorphisms, comparative genomic hybridization array and microsatellite analysis are claimed to be able to detect more subtle structural alterations [23, 28, 29, 32–36]. However, changes such as base substitutions, very small deletions or insertions are still missed [13]. Moreover, the FISH technique has been considered an appropriate and reliable method when compared to standard histopathology investigation, and it is easily available in many clinical laboratories [2, 17].

In conclusion, this study confirms that FNAB is a safe and reliable technique for sampling in-vivo uveal melanoma for monosomy 3 detection. Using the same sampling technique, we will probably be able to investigate more subtle cytogenetic and molecular characteristics of uveal melanoma. The identification of patients who have a high risk of developing metastatic disease would allow providing closer monitoring for earlier detection of metastases or adjuvant chemotherapy, currently under laboratory investigation, to prevent metastatic disease.

References

- Seregard S (1996) Posterior uveal melanoma. *Acta Ophthalmol Scand* 74:315–329
- Naus NC, Verhoeven ACA, van Drunen E, Slater R, Mooy CM, Paridaens DA, Luyten GP, de Klein A (2002) Detection of genetic prognostic markers in uveal melanoma biopsy using fluorescence in situ hybridization. *Clin Cancer Res* 8:534–539
- Gamel JW, McLean IW, McCurdy JB (1993) Biologic distinctions between cure and time to death in 2892 patients with intraocular melanoma. *Cancer* 119:786–791
- Kujala E, Kivela T (2005) Tumor, node, metastasis classification of malignant ciliary body and choroidal melanoma evaluation of the 6th edition and future directions. *Ophthalmology* 112:1135–1144
- Schmitt A, Bechrakis NE, Martus P, Mutlu D, Sheibenbogen C, Bornfeld N, Foerster MH, Thiel E, Keilholz U (2004) Independent prognostic factors for distant metastases and survival in patients with primary uveal melanoma. *Eur J Cancer* 40:2389–2395
- Gambrelle J, Labialle S, Dayan G, Gayet L, Barakat S, Grange JD, Baggetto LG (2004) Toward monosomy 3 as the main prognosis factor of uveal melanoma: current cytogenetic data. *J Fr Ophthalmol* 27:1061–1067
- Kilic E, Naus NC, van Gils W, Claver CC, van Til ME, Verbiest MM, Stijnen T, Mooy CM, Paridaens D, Beverloo HB, Luyten GP, de Klein A (2005) Concurrent loss of chromosome arm 1p and chromosome 3 predicts a decreased disease-free survival in uveal melanoma patients. *Invest Ophthalmol Vis Sci* 46:2253–2257
- Scholes AGM, Damato BE, Nunn J, Hiscott P, Grierson I, Field JK (2003) Monosomy 3 in uveal melanoma: correlation with clinical and histologic predictors of survival. *Invest Ophthalmol Vis Sci* 44:1008–1011
- Höglund M, Gisselsson D, Hansen GB, White VA, Sall T, Mitelman F, Horsman D (2004) Dissecting karyotypic patterns in malignant melanomas: Temporal clustering of losses and gains in melanoma karyotypic evolution. *Int J Cancer* 108:57–65
- Sisley K, Rennie IG, Cottam DW, Potter AM, Potter CW, Rees RC (1990) Cytogenetic findings in six posterior uveal melanomas: involvement of chromosomes 3, 6 and 8. *Genes Chromosomes Cancer* 2:205–209
- Sandinha M, Farquharson M, McKay I, Roberts F (2005) Monosomy 3 predicts death but not time to death in choroidal melanoma. *Invest Ophthalmol Vis Sci* 46:1497–1501
- Onken M, Worley L, Ehlers J, Arbour JW (2004) Genes expression profiling reveals two molecular classes and predict metastatic death. *Cancer Res* 64:7205–7209
- Prescher G, Bornfeld N, Becher R (1990) Non-random chromosomal abnormalities in primary uveal melanoma. *J Natl Cancer Inst* 82:1765–1769
- Sisley K, Cottam DW, Rennie IG, Parson MA, Potter AM, Potter CW, Rees RC (1992) Non-random abnormalities of chromosomes 3, 6, and 8 associated with posterior uveal melanoma. *Genes Chromosomes Cancer* 5:197–200
- Sisley K, Tattersall N, Dyson M, Smith K, Mudhar H, Rennie I (2006) Multiplex fluorescence in situ hybridization identifies novel rearrangements of chromosome 6, 15, and 18 in primary uveal melanoma. *Exp Eye Res* 83:554–559
- Tschentscher F, Hüsing J, Hölter T, Kruse E, Dresen IG, Jockel KH, Anastassiou G, Schilling H, Bornfeld N, Horsthemke B, Lohmann DR, Zeschnigk M (2003) Tumor classification based on gene expression profiling shows that uveal melanomas with and without monosomy 3 represent two distinct entities. *Cancer Res* 63:2578–2584
- Sisley K, Nichols C, Parsons MA, Farr R, Rees RC, Rennie IG (1998) Clinical applications of chromosome analysis, from fine needle aspiration biopsies, of posterior uveal melanomas. *Eye* 12:203–207
- Midena E, Bonaldi L, Parrozzani R, Tebaldi E, Boccassini B, Vujosevic S (2006) In vivo detection of monosomy 3 in eyes with medium-sized uveal melanoma using transscleral fine needle aspiration biopsy. *Euro J Ophthalmol* 16:422–425
- Prescher G, Bornfeld N, Hirsch H, Horsthemke B, Jockel KH, Becher R (1996) Prognostic implications of monosomy 3 in uveal melanoma. *Lancet* 347:1222–1225
- Kvanta A, Seregard S, Kopp ED, All-Ericsson C, Landau I, Berglin L (2005) Choroidal Biopsies for intraocular tumors of indeterminate origin. *Am J Ophthalmology* 140:1002–1006
- Shields JA, Shields CL, Ehya H, Eagle RCJ, De Potter P (1993) Fine-needle aspiration biopsy of suspected intraocular tumors. The 1992 Urwick Lecture. *Ophthalmology* 100:1677–1684

22. Caminal JM, Sanz S, Carreras M, Català I, Arruga J, Rocca G (2006) Epibulbar seeding at the site of a transvitreal fine-needle aspiration biopsy. *Arch Ophthalmology* 124:587–589
23. Shields CL, Ganguly A, Materin MA, Teixeira L, Mashayekhi A, Swanson LA, Marr BP, Shields JA (2007) Chromosome 3 analysis of uveal melanoma using fine-needle aspiration biopsy at the time of plaque radiotherapy in 140 consecutive cases. *Arch Ophthalmol* 125:1017–1024
24. Sen J, Groenewald C, Hiscott P, Smith P, Damato B (2006) Transretinal choroidal tumor biopsy with a 25-gauge vitrector. *Ophthalmology* 113:1028–1031
25. Sandinha T, Farquharson M, McKay I, Roberts F (2006) Correlation of Heterogeneity for chromosome 3 copy number with cell type in choroidal melanoma of Mixed-cell type. *Invest Ophthalmol Vis Sci* 47:5177–5180
26. Maat W, Jordanova ES, van Zelderen-Bhola SL, Barthen ER, Wessels HW, Schalijs-Delfos NE, Jager MJ (2007) The heterogeneous distribution of monosomy 3 in uveal melanomas: implications for prognostication based on fine-needle aspiration biopsies. *Arch Pathol Lab Med* 131:91–96
27. Meir T, Zeschnigk M, Masshofer L, Pe'er J, Chowers I (2007) The spatial distribution of monosomy 3 and network vasculogenic mimicry patterns in uveal melanoma. *Invest Ophthalmol Vis Sci* 48:1918–1922
28. Onken MD, Worley LA, Parson E, Char HD, Bowcock AM, Harbour JW (2007) Prognostic testing in uveal melanoma by transcriptomic profiling of fine needle biopsy specimens. *Clin Cancer Res* 13:2923–2927
29. Worley LA, Onken MD, Parson E, Robirds D, Branson J, Char DH, Perry A, Harbour JW (2007) Transcriptomic versus chromosomal prognostic markers and clinical outcome in uveal melanoma. *Clin Canc Res* 13:1466–1471
30. Sisley K, Rennie IG, Parsons MA, Jacques R, Hammond DW, Bell SM, Potter AM, Rees RC (1997) Abnormalities of chromosomes 3 and 8, in posterior uveal melanoma, correlate with prognosis. *Genes Chromosom Cancer* 19:22–28
31. Collaborative Ocular Melanoma Study Group (2001) Report No. 26. Development of metastatic disease after enrollment in the COMS trials for treatment of choroidal melanoma. *Arch Ophthalmol* 119:961–965
32. Aalto Y, Eriksson L, Seregard S, Larsson O, Knuutila S (2001) Concomitant loss of chromosome 3 and whole arm losses and gains of chromosome 1, 6, or 8 in metastasizing primary uveal melanoma. *Invest Ophthalmol Vis Sci* 42:313–317
33. Hughes S, Damato BE, Giddings I, Hiscott PS, Humphreys J, Houlston RS (2005) Microarray comparative genomic hybridisation analysis of intraocular uveal melanomas identifies distinctive imbalances associated with loss of chromosome 3. *Br J Cancer* 93:1191–1196
34. Tschentscher F, Prescher G, Zeschnigk M, Horsthemke B, Lohmann DT (2000) Identification of chromosome 3, 6, and 8 aberrations in uveal melanoma by microsatellite analysis in comparison to comparative genomic hybridization. *Cancer Genet Cytogenet* 122:13–17
35. Worley LA, Onken MD, Person E, Robirds D, Branson J, Char DH, Perry A, Harbour JW (2007) Transcriptomic versus chromosomal prognostic markers and clinical outcome in uveal melanoma. *Clin Cancer Res* 13:1466–1471
36. Onken MD, Worley LA, Person E, Char DH, Bowcock AM, Harbour JW (2007) Loss of Heterozygosity of chromosome 3 detected with single nucleotide polymorphisms is superior to monosomy 3 for predicting metastasis in uveal melanoma. *Clin Cancer Res* 13:2923–2927